Midterm Solution Report

Project 15: *Heart attack prediction*

By: *Luck Poyet*

1. Implemented Model So Far

1.1 Model

The first model implemented is the baseline **Logistic Regression**. To predict whether a patient actually has a *heart disease risk* or not, the `Logistic Regression` model assigns a **weight coefficient** to each patient's feature based on how strong its correlation with actual heart disease is. Then, it calculates a **weighted linear combination** of these now weighted features and passes it through the **sigmoid function** ($\sigma(z) = \frac{1}{1+e^{-z}}$), which thus transforms the linear output into a **clean probability**.

Iteration after iteration, the model passes its result through the **Loss Function** and tweaks its weights accordingly by performing **gradient-descent** of it, consistently lowering the error rate until the end of the training phase.

Moreover, the model's **hyperparameters** are being tweaked for a complete evaluation.

Parameter	Values	Description
`C`	`0.01, 1.0`	*Scales the weight values. A lower value prevents overfitting better but can hinder performance.*
`class_weight`	`none` or `balanced`	*Either doesn't or does adjust weight values inversely proportional to patient status frequencies (healthy/at risk). Prevents the model from lazily following the overall health tendency.*

1.2 Preprocessing

The **dataset preprocessing** was implemented as well. As mentioned in the *Preliminary Report*, it consists of *Median Imputation & Normalization* for the `NUMERICAL_FEATURES`, and *`Null` blank-filling method & OneHotEncoding* for the `CATEGORICAL_FEATURES`.

2. Results Analysis

With the first model implementation, a scalable output method has been designed in order to suit any upcoming model.

The method consists of the following metrics from the *sk-learn* `classification_report` method:

Precision	Among every patient the model framed as *ill* (1), how often it was right.
Recall	Among every actually *ill* (1) patient, how often they were correctly framed by the model.
f1 Score	Harmonic mean of the two previous metrics, following the formula : $2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}$

> To properly evaluate the overfitting, the program directly subtracts the *testing phase final accuracy* from the *training phase*'s. One may then consider **the lower the difference the less prone to overfitting the model is**.

2.2 Raw Data Output

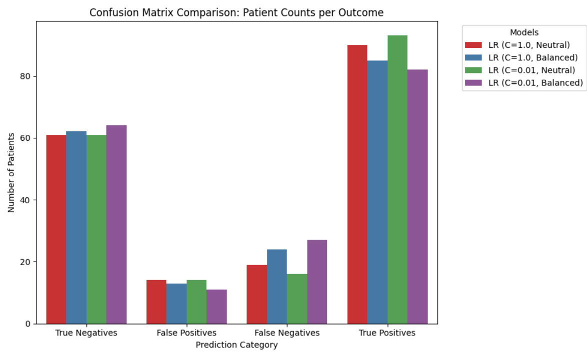
- Logistic Regression

	LR (C=1.0, Neutral)	LR (C=1.0, Balanced)	LR (C=0.01, Neutral)	LR (C=0.01, Balanced)
Test Accuracy	0.8207	0.7989	0.8370	0.7935
Overfitting Gap	0.0136	0.0367	-0.0082	0.0326
Precision (Risk)	0.8654	0.8673	0.8692	0.8817
Recall (Risk)	0.8257	0.7798	0.8532	0.7523
F1-Score (Risk)	0.8451	0.8213	0.8611	0.8119

> With only a **4%** variation in accuracy and a **sub-1%** overfitting rate, one can conclude that all the hyperparameter sets performed somewhat evenly.
> While the `precision` remains quite constant all along, with a slight rise for the 4th set, it seems that a `balanced` *class_weight* consistently results in a lower `recall` value.

2.3 Bar Graph Plotting

As seen previously, the results remain quite packed. One may still notice that the `balanced weight_class` hyperparameter, while keeping up on most of the metrics, resulted in a higher number of **False Negatives**, which is the most harmful type of error from a medical standpoint.



3. Conclusion So Far

As expected, the `Logistic Regression` model seems to be suited for such a problem so far. With these relevant yet improvable results, it acts as the perfect baseline for the following, more complex models to surpass.

Regarding the final, 3rd part of the project, several additions should be considered such as the implementation of a **K-fold cross validation** and the setup of the **ablation study**, especially on the `origin` feature which is left aside for now.

Moreover, a more synthetic visual output should be considered. While plotting every hyperparameter set output is necessary for a single model in use, selecting either the average or the best set's result per model could prevent the final diagram from getting bloated.

Such decision therefore has to be made during the next **Teams project meeting**.